

Scientific Computing

Mathematical Foundations for Machine Learning

Seminar 1: Problem Set

Probability Models, Conditioning, and Random Variables

Block I, following Lecture 1

Problem 1 (The Monty Hall protocol). Three doors. Behind one of them is an internship in a decent laboratory; behind the other two, goats. You point at a door. The host, *who knows what is behind each door*, opens one of the two remaining doors, always revealing a goat, and offers you the chance to switch. Write c for the door hiding the prize, g for the door you guessed, and h for the door the host opened.

- Construct a probabilistic model of the experiment.
- Assume you guessed door 1. Compute $\mathbb{P}(c = 1 \mid h = 3)$ and $\mathbb{P}(c = 2 \mid h = 3)$ directly from the definition of conditional probability. Should you switch?
- Identify the exact place in your computation where the assumption that the host *knows* the answer is used.

Remark. This is the question put to a student in the film *21* (2008); answering it correctly earns him a seat on a card-counting team. Evidence that a conditional probability exercise occasionally changes a life.

Problem 2 (Monty Fall). The same three doors, but this time the host does *not* know what is behind them. He trips, and in falling pushes open one of the two doors you did not choose, each with probability $\frac{1}{2}$. It happens to reveal a goat. He offers you the chance to switch.

- Compute the probability that the prize is behind your original door, given that a goat was revealed. Should you switch?
- You see exactly what you saw in Problem 1: your door closed, one other door open, a goat behind it. Explain why the answer is nevertheless different.
- Which single quantity in the model of Problem 1 has changed?

Problem 3 (A detector that is right 95% of the time). A university runs every submitted essay through an AI-text detector. Its characteristics have been validated honestly:

$$\mathbb{P}(\text{flag} \mid \text{AI}) = 0.95, \quad \mathbb{P}(\text{flag} \mid \text{human}) = 0.05.$$

The dean's office estimates that one essay in twenty is AI-generated, so the base rate is $\mathbb{P}(\text{AI}) = 0.05$. Your essay has been flagged.

- Compute $\mathbb{P}(\text{AI} \mid \text{flag})$.
- How large would the base rate $\mathbb{P}(\text{AI})$ have to be for a single flag to give a posterior probability above 0.9?

Problem 4 (Two detectors, and the odds form of Bayes' rule). The university adds a second detector with the same two characteristics, and both detectors flag your essay. Write H for the hypothesis "AI-generated", H^c for its complement, and $E_1, \dots, E_n$ for the events that detectors $1, \dots, n$ raise a flag. Assume throughout that $E_1, \dots, E_n$ are conditionally independent given the class.

- Compute $\mathbb{P}(H \mid E_1 \cap E_2)$ directly from Bayes' theorem.
- Now repeat the computation in a form that scales with n . Define the *posterior odds* and *prior odds*

$$\mathcal{O}(H \mid E) = \frac{\mathbb{P}(H \mid E)}{\mathbb{P}(H^c \mid E)}, \quad \mathcal{O}(H) = \frac{\mathbb{P}(H)}{\mathbb{P}(H^c)},$$

and the *likelihood ratio* of a single flag, $\text{LR} = \mathbb{P}(E_i | H) / \mathbb{P}(E_i | H^c)$.

Write Bayes' theorem for H and for H^c , take the ratio, and observe that the normalizing denominator — the sum over all routes to the evidence — is common to both and cancels. Using conditional independence to factor $\mathbb{P}(E_1 \cap \dots \cap E_n | H)$, deduce that for n detectors with identical characteristics

$$\mathcal{O}(H | E_1 \cap \dots \cap E_n) = \text{LR}^n \cdot \mathcal{O}(H).$$

Convert back with $\mathbb{P} = \mathcal{O} / (1 + \mathcal{O})$, check that $n = 2$ reproduces part (a), and find the smallest n for which the posterior probability exceeds 0.99.

Remark (On the assumption behind LR^n). Conditional independence of the flags given the class is a strong requirement, and in practice it is almost never met. Detectors trained on similar corpora learn similar surface statistics, so their errors are correlated *within* each class: an essay whose style happens to mislead the first detector tends to mislead the second as well. When that is so, the joint likelihood ratio of n detectors is strictly smaller than LR^n , and each additional flag carries less new information than the one before it. In the limiting case where the second detector merely reproduces the verdict of the first, it carries none at all: the joint likelihood ratio is that of a single detector, and the posterior odds after two flags are exactly the posterior odds after one. The factorized formula is therefore an upper bound, and treating correlated evidence as independent makes the posterior systematically overconfident — the usual reason why naive Bayes classifiers rank cases acceptably but are poorly calibrated.

Problem 5 (The best of several runs). A random search launches two independent training runs. The quality of each run is uniformly distributed on $\{1, 2, \dots, 6\}$ on some fixed scale, and the two runs are independent. You keep the better of the two: $X = \max(R_1, R_2)$.

- Find the CDF $F_X(k)$ for $k = 1, \dots, 6$ *before* attempting the PMF, and explain why the CDF is the easier object here.
- Recover the PMF using $p_X(k) = F_X(k) - F_X(k^-)$.
- Compute $\mathbb{P}(X \geq 5)$.
- Generalize to n independent runs. How large must n be for $\mathbb{P}(X = 6) > 0.9$?

Problem 6 (A density is not a probability, and what it is for).

- Let $f_Y(y) = 10$ for $y \in [0, 0.1]$, and let $f_Y(y) = 0$ otherwise. Verify that f_Y is a legitimate density and compute $\mathbb{P}(Y = 0.05)$.
- Let $f_X(x) = 2x$ on $[0, 1]$ and 0 elsewhere. Find F_X and its inverse F_X^{-1} on $[0, 1]$.
- Let $U \sim \text{Uniform}(0, 1)$. Show that $X = F_X^{-1}(U)$ has cumulative distribution function F_X . The proof is one line; begin with $\mathbb{P}(X \leq x)$.
- Write down the same construction for the exponential distribution with $F(x) = 1 - e^{-\lambda x}$, $x \geq 0$.